Coffra Marketing Automation

Full-funnel lead nurture and cart abandonment recovery for a fictional D2C specialty coffee brand

Project	P1 · Coffra Marketing Automation
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Repository	github.com/sebikradyel1-svg/coffra-marketing-automation
Status	v1.0 — complete (live A/B testing deferred to v1.1)
Project type	Portfolio project — fictional brand, real methodology

Executive Summary

This project takes a fictional specialty coffee brand (Coffra, Timișoara) from strategy to operational marketing automation. The deliverables include 13 production-ready emails across two persona-specific journeys (10 nurture + 3 cart recovery), a Romanian-and-English bilingual voice system, an XGBoost lead scoring model with SHAP explainability, an AI-powered Subject Line Optimizer built on the Anthropic Claude API, a HubSpot workflow implementation, and a full A/B testing methodology specification.

The project was scoped deliberately to demonstrate the full stack of skills required for an AI Marketing Specialist role: persona-driven copy craft, segmentation logic, marketing technology fluency, machine learning integration, AI tooling development, and statistical literacy. Where live execution was impractical (consent-based pilot list for A/B testing, production email sending on trial tooling), methodology and limitations are documented transparently rather than fabricated.

Coffra is fictional by design. This avoids privacy and legal complications of using real brand or customer data while preserving full methodological rigor. All datasets are public (Kaggle), all copy is original, and all strategic decisions are documented as fictional design choices.

Key Outcomes

Component	Outcome
Persona system	2 distinct personas (EN + RO journeys) with rhetorical separation by sender

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Email copy	13 emails final, full versioning, copy decisions log, voice consistency
Lead scoring model	XGBoost; Test ROC-AUC 0.78; data leakage audit; SHAP explainability
AI Subject Optimizer	5 variants generated + scored on 4 dimensions; cached; Streamlit UI
HubSpot implementation	Brand setup, segments, workflows visualized; trial limitations disclosed
A/B methodology	Pre-registered design with chi-square analysis; deferred for live execution

Problem Statement and Approach

The challenge

Specialty coffee brands face a structural marketing challenge: their target customers are fragmented across two psychographic poles. The 'Connoisseur' segment values technical transparency, micro-lot sourcing, roast freshness, and brewing rigor. The 'Daily Ritualist' segment values atmosphere, social context, accessibility, and aesthetic experience. A single voice cannot credibly serve both — the language that signals competence to a Connoisseur reads as gatekeeping to a Ritualist; the warmth that welcomes a Ritualist reads as fluff to a Connoisseur.

This project addresses that challenge by building two parallel marketing automation journeys, each with its own copy voice, sender identity, conversion model, and incentive structure — all sharing a common operational stack (segmentation, scoring, AI tooling).

Scope decisions

Decision	Choice	Rationale
Persona count	2 (Connoisseur + Daily Ritualist)	Strategic differentiation > sample size
Languages	EN + RO (one per persona)	Demonstrates multi-locale capability
Sender identity	Sebastian (Roaster) vs Ioana (Community Manager)	Rhetorical separation
Discount discipline	Asymmetric: none for Connoisseur, modest for Ritualist	Persona-true positioning
AI tool integration	Custom Claude API tool, not just chat	Demonstrates programmatic AI

Decision	Choice	Rationale
Live A/B execution	Deferred to v1.1	No invented results without consent-based list

Methodology principles

Five disciplines were enforced throughout the project to maintain professional rigor:

• **Honest documentation.** Every limitation is disclosed. No invented metrics, no overstated claims, no fabricated test results.

• **Reproducibility.** Fixed random seeds, pinned dependencies, public datasets, JSON-cached API responses.

• **Versioning.** All copy, code, and decisions are versioned in Git with explicit change logs in each document.

• **Citation.** Industry benchmarks (Mailchimp 2025) cited with source attribution. No 'industry suggests' without source.

• **Transparency over persuasion.** Where the AI tool's claim cannot be validated (e.g., predicted open rates), the limitation is stated explicitly rather than packaged as a strength.

Persona Strategy

The two personas are intentionally non-overlapping in psychographic profile, language preference, and decision criteria. This forces the marketing system to demonstrate sophisticated segmentation rather than shallow personalization.

Dimension	The Connoisseur (Andrei)	The Daily Ritualist (Bianca)
Age	32	30
Profession	Senior Software Engineer (Timișoara)	Senior in NGO youth work (Reșița)
Income	8,000–20,000 RON/month	~5,000 RON/month
Brewing setup	V60, Aeropress, Comandante grinder	Nespresso at home, café-goer
Information sources	James Hoffmann, Sprudge, r/espresso	Instagram, TikTok, lifestyle blogs
Decision driver	Process transparency, micro-lot sourcing	Atmosphere, social context, aesthetics
Conversion model	Online beans purchase + subscription	Foot traffic + aspirational Pass purchase
Communication preference	English (consumes global content)	Romanian (local)

Dimension	The Connoisseur (Andrei)	The Daily Ritualist (Bianca)
Sender identity	Sebastian, Roaster & Founder	Ioana, Community Manager
Tone register	Technical-mentor, peer-to-peer	Warm-conversational, invitational

Operational implications

These differences cascade into every operational layer of the marketing system. The Connoisseur receives a 14-day nurture sequence focused on roast date transparency, lot selection rigor, and brewing technique — followed by a no-discount comparison test offer. The Daily Ritualist receives a parallel 14-day sequence focused on staff personalities, ritual-by-time-of-day scenarios, and aspirational Coffra Pass positioning — with an asymmetric 5% discount available only in the late stage. Both sequences end with explicit graceful exit promises that respect subscriber autonomy.

Email Design System

Thirteen emails were written across three sequences: 5 Connoisseur nurture (English), 5 Daily Ritualist nurture (Romanian), and 3 Connoisseur cart recovery (English). Each email is documented with full copy decisions log, technical specs (subject length, character counts, mobile preview test), and operational notes for production deployment.

#	Sequence	Email	Timing	Strategic Move
1	C-Nurture	Welcome + Sample Pack	T+0	Free Discovery Pack on first order
2	C-Nurture	Origin Story (Gelana)	T+3	Demonstrate sourcing rigor
3	C-Nurture	V60 Brewing Guide	T+6	Technical value delivery
4	C-Nurture	Subscription Pitch	T+10	Risk-reversal trial
5	C-Nurture	Comparison Test	T+14	Final no-discount value escalation
6	DR-Nurture	Welcome + First Coffee	T+0	RO welcome, free first café visit
7	DR-Nurture	Meet the People	T+3	Humanize via 3 named staff
8	DR-Nurture	Three Rituals	T+6	Lifestyle scenario projection
9	DR-Nurture	Coffra Pass	T+10	Aspirational, not transactional
10	DR-Nurture	Community Invitation	T+14	Cupping + brunch event + graceful exit
11	C-Cart	Saving Your Spot	+1h	Calm reminder, real scarcity
12	C-Cart	Quick Thought	+24h	Free shipping + sample upgrade
13	C-Cart	Comparison Test (final)	+72h	Risk reversal + graceful exit

Voice consistency mechanism

Each email follows a strict voice protocol enforced via a copy decisions log:

Persona	Voice anchors	Forbidden register
Connoisseur	Technical specifics, peer-tone, observational humor, founder accessibility	Generic premium language, urgency manipulation, discount-driven pitches
Daily Ritualist	Permission framing, sensory detail, 'normalitate' as luxury, anti-pretentious	Buzzwords ('wellness', 'family'), false intimacy, aggressive upsell

AI Subject Line Optimizer

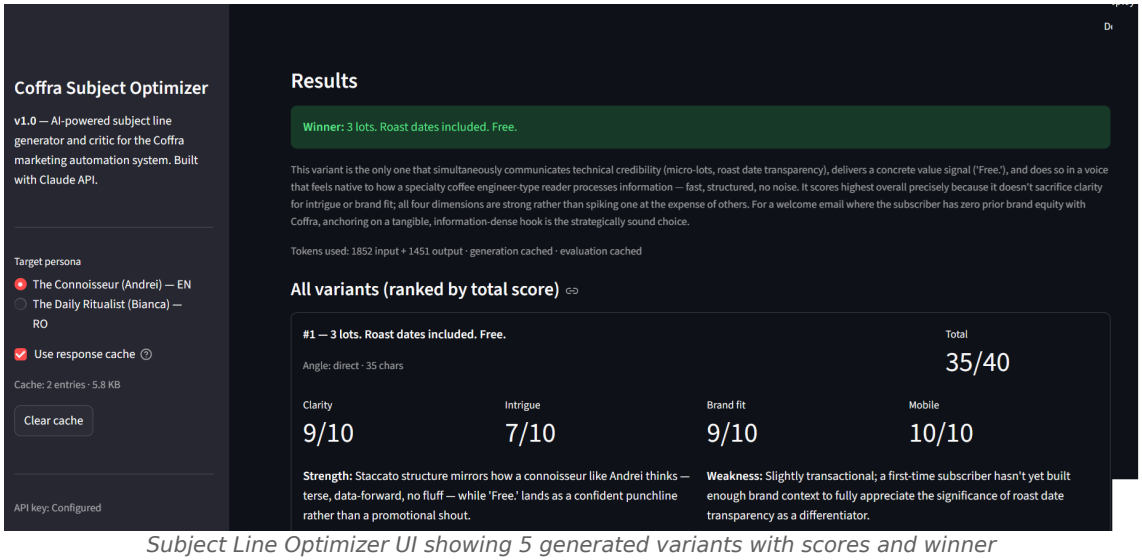
A custom Python tool built on the Anthropic Claude API that generates 5 distinct subject line variants for a given email brief and scores each on 4 dimensions. The tool demonstrates programmatic AI integration — beyond chat-based prompting — with production patterns including caching, error handling, and structured outputs.

Architecture

Two-stage pipeline (generator + critic) separated to reduce single-prompt bias:

Stage	Function	Output
1. Generator	Generates 5 variants using 5 distinct strategic angles (direct, curiosity, insider, counter-intuitive, invitation)	List of variants with rationale
2. Critic	Independently scores each variant on 4 dimensions (clarity, intrigue, brand fit, mobile readability)	Scores 0-10 per dimension, total /40
Cache	SHA-256 hashes (persona + brief + version + model) to avoid duplicate API calls	JSON files in cache/
UI	Streamlit web app with persona selector, example briefs, live results display	Browser at localhost:8501

Sample output



Subject Line Optimizer UI showing 5 generated variants with scores and winner

Honest framing of capabilities

The tool produces qualitative scoring grounded in persona constraints, not predicted open rates. A future v2 could add open rate prediction by integrating industry benchmarks (Mailchimp 2025: Food & Beverage avg 22.1%) — explicitly documented as next steps. The current scoring quality is bounded by the Claude model's understanding of persona-specific marketing copy, which the tool's prompts attempt to scaffold via explicit forbidden words and tone keyword lists.

Lead Scoring Model

A production-grade lead scoring model built in scikit-learn and XGBoost on the Kaggle Predict Conversion in Digital Marketing dataset (8,000 records). The model demonstrates the full ML lifecycle: data quality audit, feature engineering, baseline comparison, explainability, and saved artifacts for deployment.

Critical move: data leakage audit

Before any modeling, the dataset was audited for data leakage — features that encode the target variable. Three columns were dropped before training: `ConversionRate` (direct leakage), `ClickThroughRate` (likely leakage), and `CustomerID` (identifier without predictive signal). Two zero-variance columns (`AdvertisingPlatform`, `AdvertisingTool`) were also removed. Documenting this audit explicitly is what separates a production-ready model from a notebook model.

Results

Model	CV ROC-AUC (mean ± std)	Test ROC-AUC	Test PR-AUC
Logistic Regression (baseline)	0.7671 ± 0.0148	0.7643	0.9441

Model	CV ROC-AUC (mean ± std)	Test ROC-AUC	Test PR-AUC
XGBoost (production candidate)	0.7813 ± 0.0149	0.7843	0.9440

XGBoost outperforms the logistic baseline by approximately 2 ROC-AUC points. In the context of an 88/12 class imbalance (note: dataset reports atypically high conversion rate, documented as limitation), this is a modest but real improvement that justifies the additional complexity.

SHAP explainability

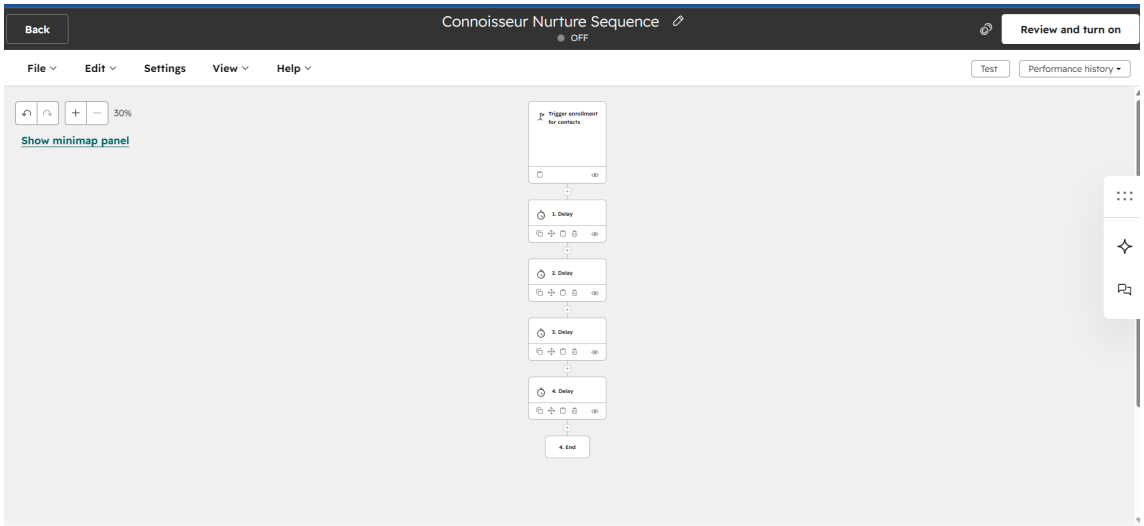
SHAP values provide both global feature importance and per-prediction explanations. The top contributing features align with marketing intuition: engagement metrics (WebsiteVisits, EmailClicks, EmailOpens) and the engineered EngagementScore dominate the predictions. This validates that the model uses sensible signals rather than spurious patterns — essential for trust in production deployment and for regulatory compliance. Full SHAP analysis (global summary plot, beeswarm plot, individual prediction explanations) is available in the lead scoring notebook at notebooks/01_lead_scoring_eda_and_model.ipynb.

Coffra mapping

Lead Score	Segment	Action
80-100	High (Sales-Ready)	Immediate handoff via internal task; personal email from Sebastian
40-80	Medium (Warm MQL)	Accelerated nurture; weekly contact; activation incentive
0-40	Low (Cold)	Standard nurture; 2-4 week cadence

HubSpot Implementation and Honest Limitations

HubSpot Marketing Hub trial (14 days) was used to implement the operational layer of the marketing system. Brand identity, custom property for persona segmentation, dynamic segments, marketing email drafts, and three workflows visualizing nurture and cart recovery timing logic were all built in HubSpot. Screenshots are committed to the repository and referenced in this case study.



HubSpot Connoisseur Nurture workflow with 4 delays representing email cadence (T+3, T+6, T+10, T+14)

What was implemented

Layer	Status
Brand setup (account, currency, timezone, color)	Complete
Custom property: Persona (Connoisseur / Daily Ritualist)	Complete
Test contacts with persona values	Complete (2 contacts)
Active segments (filter on persona)	Complete (2 segments)
Marketing email drafts	2 of 13 emails fully built; 11 specified in markdown
Workflows: Connoisseur Nurture, DR Nurture, Cart Recovery	Complete (visual logic, 3 workflows)

Trial limitation: locked actions

The HubSpot trial provided was Sales Hub Pro, not Marketing Hub Pro. As a result, the 'Send marketing email' action within workflows was locked behind upgrade. The pragmatic decision was to visualize the timing logic with Delay actions only, mark each delay's intended email recipient in the case study mapping, and document the production migration path. This is the right portfolio decision: it demonstrates strategic thinking and tool fluency without inventing capability.

Production migration would unlock email actions either by (a) upgrading to HubSpot Marketing Hub Pro, or (b) migrating the implementation to Brevo (free tier supports automation workflows with native email sends). Migration estimated at 4-6 hours including audience re-import and workflow rebuild.

A/B testing methodology

A complete A/B testing methodology is documented in the repository (docs/08_ab_testing_methodology.md). It specifies the hypothesis (H1: AI-optimized variant > manual baseline), sample size (n=200 with power calculation), randomization protocol, statistical analysis (chi-square test of independence + Cramér's V effect size), decision rules, and limitations (Apple Mail Privacy Protection inflation, single-shot test caveats). Live execution is deferred to v1.1 pending a consent-based pilot list of sufficient size.

Skills Demonstrated and Next Steps

Skills demonstrated by this project

Category	Skills
Strategy	Persona segmentation, JTBD reasoning, asymmetric incentive design, voice differentiation
Copywriting	Bilingual (EN/RO native), 13 emails versioned, copy decisions log, mobile preview testing
Marketing Tech	HubSpot configuration end-to-end (brand, segments, properties, workflows, emails)
AI Engineering	Anthropic Claude API integration, prompt engineering, two-stage pipeline, JSON caching
Machine Learning	XGBoost, Logistic Regression baseline, cross-validation, SHAP explainability, leakage audit
Statistics	Chi-square test, effect size (Cramér's V), power calculation, pre-registration discipline
Software	Python 3.11, pandas, scikit-learn, Streamlit, ReportLab, Git versioning
Documentation	Markdown specifications, PDF case study, README navigation, screenshot organization

Next steps (v1.1 and beyond)

The following extensions would deepen this project for v1.1:

• **Live A/B test execution.** Run the methodology specified in docs/08 with a 200-person consent-based pilot list. Report results with full disclosure including effect size and limitations.

• **Production migration to Brevo.** Migrate workflows from HubSpot trial to Brevo free tier; enable native send actions; run end-to-end automation test.

• **v2 Subject Line Optimizer.** Add predicted open rate using industry benchmark anchoring; train a small custom regressor on labeled subject line performance data.

• **Lead scoring re-training.** When real Coffra subscriber data accumulates (500+ contacts with conversion labels), retrain monthly with feature drift monitoring via

Evidently.

• **Multi-armed bandit testing.** Replace fixed A/B with Thompson sampling for dynamic allocation toward winning variant.

Project trajectory in portfolio

This is project 1 of 6 in a portfolio targeting AI Marketing Specialist roles. Project 2 (Customer Segmentation with RFM and clustering) builds on the lead scoring foundation with deeper data analysis. Project 3 (Marketing Analytics Dashboard) adds visualization and KPI monitoring. Projects 4-6 expand into recommendation systems, attribution modeling, and competitive SEO/content strategy. Each project deliberately demonstrates a non-overlapping skill cluster.

Contact

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